Mathieu Dabrowski

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EDUCATION

2024 – 2027 (expected) : Ph.D. in mathematical physics, LAPTH, Université Savoie Mont Blanc, Annecy, France

- Thesis : *Set-theoretic and algebraic approaches to integrable Markov processes*
- Supervisors : Eric Ragoucy (LAPTH, Annecy) and Loïc Poulain d'Andecy (Laboratoire de Mathématiques de Reims)
- Expected defence : summer 2027

2021 – 2024 : Master in physics, École Normale Supérieure de Lyon, France

- 2023 – 2024 : Long Research Project (two 5-month research internships, see "Previous research experience")
- 2022 – 2023 : M2 Physics & Chemistry : Physics, Concepts and Applications
- 2021 – 2022 : M1 Physics & Chemistry

RESEARCH INTERESTS

Integrable out-of-equilibrium Markov processes and the algebraic objects underlying such systems. In particular, integrable exclusion Markov processes generalizing the Symmetric Simple Exclusion Process (SSEP), the Matrix Ansatz method, set-theoretical solutions to the Yang-Baxter equation.

PUBLICATIONS

1. M. Dabrowski, L. Poulain d'Andecy and E. Ragoucy, *Twisted symmetric exclusion processes and set-theoretical R-matrices*, J. Phys. A: Math. Theor. **59**, 245203 (2026). arXiv:2602.18204, doi:10.1088/1751-8121/ae751a.
2. M. Dabrowski, L. Poulain d'Andecy and E. Ragoucy, *Integrable multi-species SSEP with reactive particle species*, J. Stat. Mech. (2026), 113102. arXiv:2607.18959, 10.1088/1742-5468/aeb496.
3. In preparation : M. Dabrowski, L. Poulain d'Andecy and E. Ragoucy, *Effective Ising model in an open (p, 1)-SSEP.*

TALKS, CONFERENCES & SCHOOLS

- *Sept. 2026 :* Conference RAQIS'26 (Recent Advances in Quantum Integrable Systems), Annecy, France – *talk*
- *Aug. 2026 :* Young Researchers School "Integrability and two-dimensional CFT", Budapest, Hungary – *poster*
- *June 2026 :* IMJ-PRG summer school "Integrable combinatorics", Paris, France – *poster*
- *Aug. 2025 :* Les Houches School of Physics "Exact Solvability and Quantum Information", France – *poster*

TEACHING

2025 – 2026 : Teaching assistant in mathematics, IUT d'Annecy, Université Savoie Mont Blanc

- Tutorials in mathematics for first-year B.U.T. students (Networks and Telecommunications)– 63 hours

PREVIOUS RESEARCH EXPERIENCE

Feb. – Jul. 2024 : Ghent University, Belgium, with J. Van der Jeugt – *Lie algebras and symmetric polynomials in physical models.*

Sept. 2023 – Feb. 2024 : Institut de Mathématiques de Bourgogne, Dijon, with S. Carrozza and L. Lionni – *Multipartite entanglement in random tensor networks.*

Mar. – Jun. 2023 : Laboratoire Univers et Particules de Montpellier, with J. Lavalle – *Dark matter and the primordial power spectrum on small scales with 21cm cosmology.*

2021, 2022 : Short internships in condensed matter at CEA Grenoble (theory of superconductivity, PHELIQS; experimental study of skyrmions, Spintec).

COMPUTER SKILLS

Maple, Python, AI tools (Claude).

LANGUAGES

French (native), English (C1, *Certificate in Advanced English*), Spanish (A2), Russian (beginner).

OTHER INTERESTS

Piano, chess.